SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

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Experiment - 1

Aim: To analyse human actions in a video using pose estimation techniques **Software**

Tools Required: VS Code, Python

Theory:

Human Activity Recognition (HAR) using pose estimation involves detecting and analyzing body movements in video frames to identify actions like walking, sitting, or running.We used **MediaPipe Pose** and **YOLOv8** for accurate person detection in each frame.

The combination of YOLOv8 and MediaPipe improves accuracy and efficiency by applying pose estimation only on detected humans. This approach enables real-time, multi-person tracking, and is useful in surveillance, fitness, healthcare, and gesture-based applications.

- **Pose Estimation using MediaPipe**
  MediaPipe, developed by Google, is a cross-platform framework for building multimodal ML pipelines. Its Pose solution estimates 33 2D landmarks of the human body, including the face, torso, arms, and legs.

- **Object Detection using YOLOv8**
  YOLO (You Only Look Once) is a family of real-time object detection models. YOLOv8, developed by Ultralytics, offers fast and accurate detection across multiple classes.

Code: 1A

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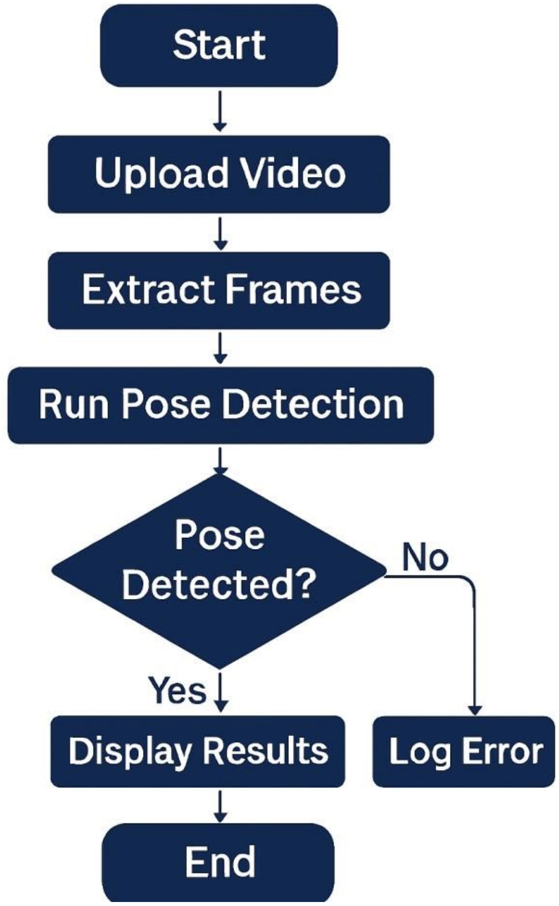
```python
#Experiment 1: Human Activity Recognition in Video

# ALGORITHM
# 1.Import libraries and load YOLOv8 and MediaPipe Pose models.
# 2.Open the video and set target frame size.
# 3.For each frame:
#   a.Resize to 1920x1080.
#   b.Detect people using YOLOv8.
#   c.For each detected person:
#      ->Crop the person region.
#      ->Run MediaPipe Pose on the crop.
#      ->Draw pose landmarks and connections on the frame.
# 4.Display the result.


import cv2
import numpy as np
from ultralytics import YOLO
import mediapipe as mp

# Load YOLOv8 model (downloads automatically if not present)
model = YOLO('yolov8n.pt')  # or 'yolov8s.pt', 'yolov8m.pt', etc.

cap = cv2.VideoCapture('Walk_2.mp4')
mp_pose = mp.solutions.pose
mp_drawing = mp.solutions.drawing_utils
pose = mp_pose.Pose(static_image_mode=False)  # Create once, outside the loop

TARGET_WIDTH = 800
TARGET_HEIGHT = 600

while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        break

    frame = cv2.resize(frame, (TARGET_WIDTH, TARGET_HEIGHT), interpolation=cv2.INTER_AREA)
    results = model(frame)
    boxes = results[0].boxes.xyxy.cpu().numpy()
    classes = results[0].boxes.cls.cpu().numpy()

    for box, cls in zip(boxes, classes):
        if int(cls) == 0:  # Class 0 is 'person' in COCO
            x1, y1, x2, y2 = map(int, box)
            person_roi = frame[y1:y2, x1:x2]
            if person_roi.size == 0:
                continue
            person_rgb = cv2.cvtColor(person_roi, cv2.COLOR_BGR2RGB)
            pose_results = pose.process(person_rgb)
            if pose_results.pose_landmarks:
                for lm in pose_results.pose_landmarks.landmark:
                    cx = int(lm.x * (x2 - x1)) + x1
                    cy = int(lm.y * (y2 - y1)) + y1
                    cv2.circle(frame, (cx, cy), 3, (0, 255, 0), -1)
                mp_drawing.draw_landmarks(
                    frame[y1:y2, x1:x2], pose_results.pose_landmarks, mp_pose.POSE_CONNECTIONS
                )

    cv2.imshow('YOLOv8 + MediaPipe Pose', frame)
    if cv2.waitKey(10) & 0xFF == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```

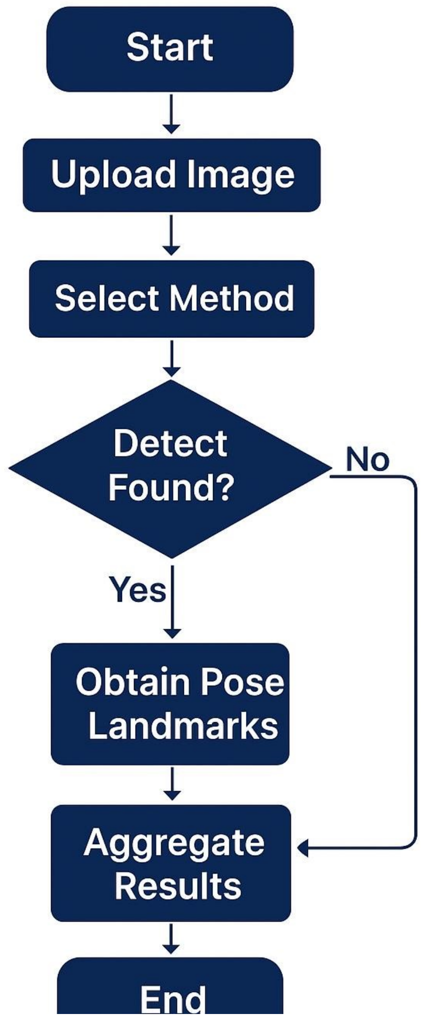
1B:

```python
# Experiment 1: Human Activity Recognition in Video
import cv2
import mediapipe as mp

mp_pose = mp.solutions.pose
mp_drawing = mp.solutions.drawing_utils
pose = mp_pose.Pose()
cap = cv2.VideoCapture('Walk.mp4')

while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        break
    frame_rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
    results = pose.process(frame_rgb)
    if results.pose_landmarks:
        landmarks = results.pose_landmarks.landmark
        from mediapipe.framework.formats import landmark_pb2
        body_landmarks = landmark_pb2.NormalizedLandmarkList()
        # Indices of body landmarks (excluding face: 0-10)
        body_indices = list(range(11, len(landmarks)))
        for idx in body_indices:
            body_landmarks.landmark.append(landmarks[idx])
        # Filter connections to only those between body landmarks
        body_connections = [
            (i-11, j-11)
            for (i, j) in mp_pose.POSE_CONNECTIONS
            if i >= 11 and j >= 11
        ]
        mp_drawing.draw_landmarks(
            frame,
            body_landmarks,
            body_connections
        )
    cv2.imshow('Pose Detection', frame)
    if cv2.waitKey(10) & 0xFF == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```

Flowchart : 1A

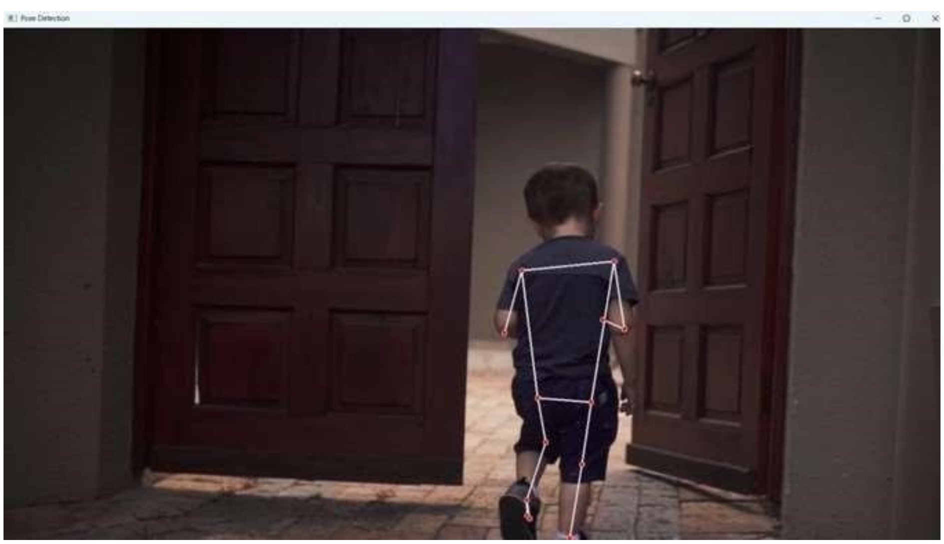

Start
Upload Video
Extract Frames
Run Pose Detection
Pose Detected?
No
Yes
Display Results
Log Error
End

1B:


Start
Upload Image
Select Method
Detect Found?
No
Yes
Obtain Pose Landmarks
Aggregate Results
End

Output: 1A:

1B:



Conclusion:

We have implemented the human action analysis in video using pose stimation techniques. By combining YOLOv8 for person detection and MediaPipe Pose for body landmark estimation, we achieved efficient and accurate multi-person pose tracking. This approach enables real-time human activity recognition and can be effectively used in applications like surveillance,healthcare monitoring, and fitness analysis.